

Vision-Language Research: Action-Conditioned Frame Prediction

Project Pipeline Formulation & Methodology

1 Problem Formulation

Let $I_t \in \mathbb{R}^{H \times W \times 3}$ denote an anchor frame sampled at discrete time t from a continuous video sequence $\mathcal{V}$, representing the static visual geometry and semantic state of a scene. Let $A = (w_1, w_2, \dots, w_L)$ denote a natural-language action directive where each token w_i belongs to a vocabulary $\mathcal{V}_{\text{text}}$, explicitly describing the causal, spatial, or dynamic transformation occurring over the time interval $(t, t + 1]$.

The objective of the proposed pipeline is to learn a conditional generative mapping:

$$\mathcal{F}_\theta : (I_t, A) \rightarrow \hat{I}_{t+1}$$

such that the synthesized frame $\hat{I}_{t+1} \in \mathbb{R}^{H \times W \times 3}$ satisfies two competing constraints:

1. **Semantic & Spatial Invariance:** Global background elements, ambient lighting, stationary objects, and unacted-upon entities in I_t must be preserved without identity drift or spatial distortion.
2. **Action-Governed Dynamics:** Acted-upon entities must transition physically and causally to conform to the precise dynamic transformation specified by A .

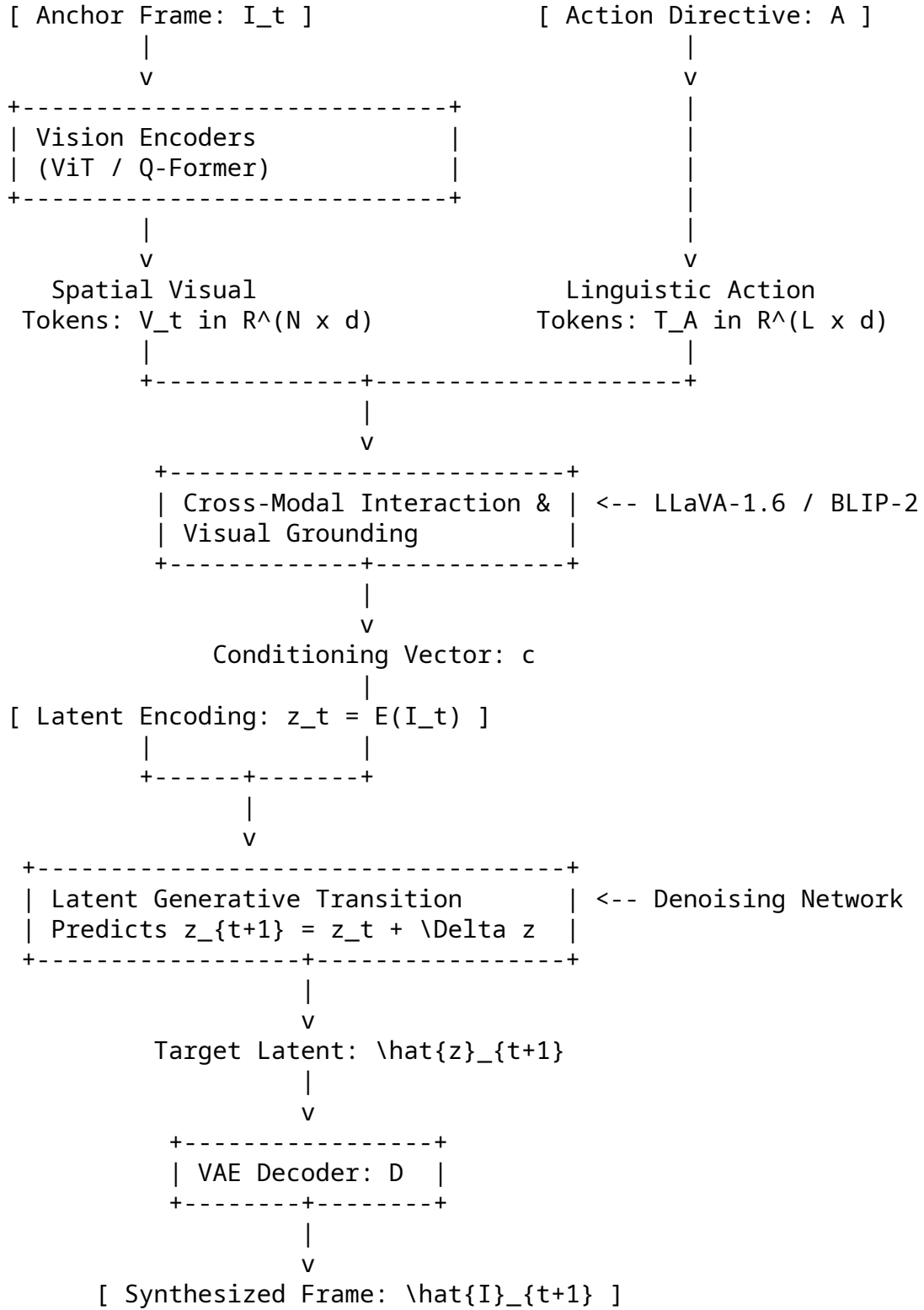
2 Core Architectural Hypothesis

Standard next-frame prediction methods either predict I_{t+1} autoregressively from prior pixel frames (suffering from compounding blur and spatial drift) or rely purely on text-to-video generation (which struggles to anchor strictly to an exact initial conditioning image without corrupting identity).

The proposed framework solves this by formulating frame prediction as a **directed visual state transition in latent space**, where vision-language foundation models (LLaVA-1.6, BLIP-2, and CLIP) act as semantic bridges. Rather than treating A as an independent global text prompt, the pipeline grounds the action A directly into the spatial coordinates of I_t , determining *which* visual entities must deform, displace, or interact, and *which* regions must remain static.

3 End-to-End Pipeline Formulation

The proposed pipeline is structured into three interdependent modules: **Multimodal Conditioning & Grounding**, **Latent State Transition**, and **High-Fidelity Frame Synthesis**.



3.1 Module A: Dual-Stream Vision-Language Conditioning

To condition generation on both visual context and textual dynamics, the pipeline decouples representation extraction into complementary streams:

- **Dense Spatial Encoding:** The anchor frame I_t is mapped into a grid of spatial feature

tokens $V_t = f_v(I_t) \in \mathbb{R}^{N \times d}$ via a Vision Transformer backbone (e.g., CLIP-ViT-L/14 or EVA-CLIP), preserving patch-level spatial layout and localized object representations.

- **Linguistic Action Encoding:** The textual directive A is mapped into sequence embeddings $T_A = f_\ell(A) \in \mathbb{R}^{L \times d}$, extracting the syntactic subject-verb-object dynamics.
- **Cross-Modal Grounding & Fusion:** To bind linguistic verbs to specific image patches, the pipeline integrates representations derived from **BLIP-2 (Q-Former)** and **LLaVA-1.6**. A set of learnable query tokens attends to both V_t and T_A through bidirectional cross-attention layers, producing a fused multimodal conditioning context:

$$c = \text{CrossAttention}(Q, [V_t \parallel T_A])$$

This ensures that the representation c explicitly links the dynamic verb in A to the spatial bounding region of the interacting entity in I_t .

3.2 Module B: Action-Conditioned Latent Transition

Operating directly in high-dimensional pixel space $\mathbb{R}^{H \times W \times 3}$ is computationally inefficient and prone to high-frequency artifacts. The pipeline projects I_t into a low-dimensional latent space via a pre-trained variational autoencoder (VAE) encoder $\mathcal{E}$:

$$z_t = \mathcal{E}(I_t), \quad z_t \in \mathbb{R}^{h \times w \times c}$$

The core transition model is formulated within a latent conditional framework:

1. **Residual Formulation:** Instead of generating z_{t+1} from pure white noise, the transition is modeled as an action-driven perturbation over z_t :

$$z_{t+1} = z_t + \Delta z(z_t, c)$$

where Δz represents the localized dynamic shift dictated by action context c .

2. **Conditioning Injection:** At each step of the iterative reverse process, the anchor latent z_t is channel-concatenated with the noisy intermediate latent z_{t+1}^τ , while the grounded multimodal representation c is injected via spatial cross-attention layers in the denoiser ϵ_θ :

$$\epsilon_\theta([z_{t+1}^\tau \parallel z_t], \tau, c)$$

This architectural split forces the model to draw stationary background pixels directly from z_t while steering moving components toward c .

3.3 Module C: Image Reconstruction & Decoding

Once the predicted latent representation $\hat{z}_{t+1}$ is synthesized, it is passed through the VAE decoder $\mathcal{D}$:

$$\hat{I}_{t+1} = \mathcal{D}(\hat{z}_{t+1})$$

The final frame is rendered at full spatial resolution, capturing fine pixel details, lighting coherence, and topological modifications governed by the action.

4 Theoretical Mechanisms Addressing Failure Modes

The proposed pipeline explicitly addresses three distinct multimodal transition challenges:

- **Mitigating Spatial Identity Drift:** In naive autoregressive setups, background details mutate over time. Concatenating z_t directly along the channel dimension of the transition backbone provides a structural skip-connection for high-frequency spatial features that should not change.
- **Resolving Dynamic Ambiguity:** A single text prompt (e.g., *"the person turns"*) has multiple valid trajectory realizations. By fusing patch-level tokens V_t with action tokens T_A , the pipeline restricts the plausible solution space to physical motions that are causally feasible given the exact starting pose in I_t .
- **Causal Disentanglement:** Using vision-language projections ensures the model can differentiate between static entity descriptors (e.g., *"red cup"*) and action directives (e.g., *"is placed on the table"*), preventing the model from repainting the entire scene when only a localized object state changes.